Adam Shebani

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EDUCATION

University of Wisconsin-Madison, College of Engineering **Madison, WI**
Degree: B.S. Computer Engineering September 2025 – December 2027
Cumulative GPA: 3.9

Madison College **Madison, WI**
General Education: Pre-Engineering August 2023 – May 2025
Cumulative GPA: 3.8

PROJECTS

- Designed and verified a fully synthesizable 16-bit CPU in SystemVerilog featuring a custom ISA, control, and datapath, validated by a 33-case directed testbench achieving full functional coverage.
- Built a full-stack web application, deployed on Railway, with a Java/Python backend supporting blog publishing, project showcases, and machine learning workflows.
- Developed a trading dashboard and a suite of quantitative research and trading tools using the Interactive Brokers API.
- Designed a 4-layer SFP+ carrier board in Altium with impedance-controlled differential pairs routed to coaxial SMA connectors for signal readout.

ENGINEERING & WORK EXPERIENCE

ECE 551 – Digital System Design & Synthesis (SystemVerilog) **UW-Madison**

- Designed and synthesized digital control logic in SystemVerilog using Synopsys Design Compiler.
- Applied timing constraints and analyzed synthesis reports to meet digital design specifications.
- Developed self-checking ModelSim testbenches and debugged issues using waveform analysis.
- Integrated an autonomous maze-solving SoC consisting of UART communication, inertial sensor interfaces, PID motor control, and FSM-based navigation in synthesizable SystemVerilog.

ECE 353 – Embedded Systems and Microcontrollers (C) **UW-Madison**

- Developed embedded C firmware on an ARM Cortex-M (PSoC 6) using interrupt-driven design with timers, GPIO, PWM, and ADC peripherals.
- Built concurrent I/O functionality using FreeRTOS to coordinate sampling, communication, and display updates without polling.
- Utilized serial communication protocols such as SPI, I2C, and UART.

CS 544 – Big Data Systems (Python) **UW-Madison**

- Built distributed storage and analytics pipelines using Docker, HDFS, Spark, Cassandra, Kafka, and gRPC.
- Developed Python data processing workflows using PySpark and Parquet.
- Designed relational schemas and RESTful backend services for distributed applications.

Badger Brothers Moving **Madison, WI**
Moving Specialist May 2025 – October 2025

- Handled customer service and on-site labor, worked full-time, completed over 100 moves, and contributed to six figures in revenue.

SKILLS

- **C, Python, Java, SystemVerilog**
- **UART, SPI, I2C, PWM, FreeRTOS**
- **Synopsys DC, Quartus, ModelSim**
- **Git, Linux, Windows**
- **Docker, HDFS, Spark**
- **Altium, KiCad, SPICE**